

# Directional structure, terrain and equipment

Version: 14/09/2026

Thesis Sections 3.5.1, 3.5.2 and 3.5.3

Thesis §3.5.1

**East–north balance in the retained archive**

## The ratio compares motion along east and north

$$\rho_{EN}(t) = \frac{\langle E_f(t)^2 \rangle_f}{\langle N_f(t)^2 \rangle_f}, \quad \langle E_f^2 \rangle_f = \text{Var}_f(E_f) + \langle E_f \rangle_f^2.$$

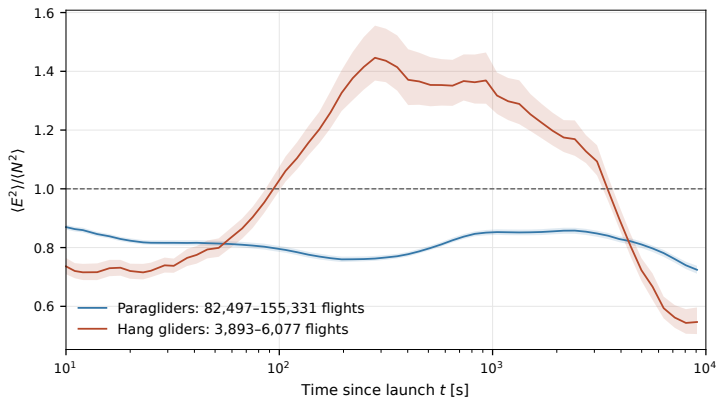
$E_f(t)$  and  $N_f(t)$  are positions relative to launch. The clock is time since launch; each time uses the flights that still contribute paired observations.

$\rho_{EN} > 1$ : larger east second moment.       $\rho_{EN} < 1$ : larger north second moment.

The ratio combines mean displacement and spread. Equality is one necessary condition for isotropy; cross moments and the full angular distribution also matter.

Thesis Section 3.5.1.

## The retained archive has unequal east and north moments



Curve ranges over  
10–10,000 s:

Paragliders: 0.72–0.87.

Hang gliders: 0.54–1.45.

The paraglider curve stays  
below one. The hang-glider  
curve crosses it.

Thesis Fig. 3.20, complete figure. Numbers are extrema of the observed curves; shading is the bootstrap range.

## The bands resample whole take-off sites and days

A site–day groups flights with the same recorded take-off site and date. Flights in one group may share weather and route conditions.

1. Resample whole groups with replacement, keeping their flights together.
2. Recompute the paired east/north moment ratio at every supported time.
3. Repeat 200 times; retain the 10th and 90th percentiles.

The shading contains the central 80% of bootstrap estimates at each time. Plots require 30 flights and 10 contributing groups.

Thesis Fig. 3.20 and the uncertainty convention used in Figs. 3.24–3.26.

Thesis §3.5.2

## **Regional principal axes**

## PCA measures spread around the mean displacement

$$\mathbf{X}_f(t, \tau) = \mathbf{r}_f(t + \tau) - \mathbf{r}_f(t), \quad \boldsymbol{\mu}_g = \langle \mathbf{X} \rangle_g,$$

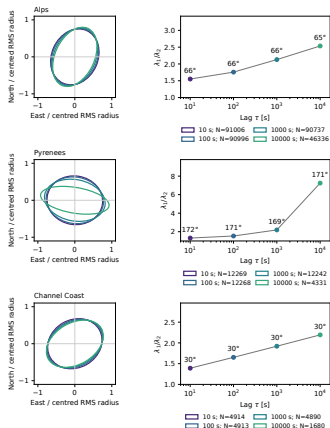
$$\Sigma_g(\tau) = \left\langle (\mathbf{X} - \boldsymbol{\mu}_g)(\mathbf{X} - \boldsymbol{\mu}_g)^\top \right\rangle_g = U_g \text{diag}(\lambda_1, \lambda_2) U_g^\top.$$

The leading eigenvector gives the axis of greatest spread;  $\lambda_1 / \lambda_2$  measures covariance anisotropy.

Use every supported increment in the regional archive on the 10-s grid. Origins are spaced by the lag within each segment. Flights with more origins receive more weight.

Thesis Section 3.5.2. Lags: 10, 100, 1000 and 10,000 s.

# The ellipses show shape and direction at four lags



Both coordinates are divided by the same centred RMS radius:

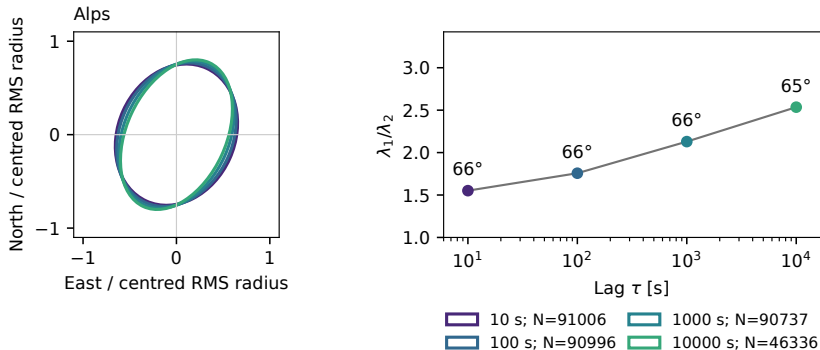
$$s_g(\tau) = \sqrt{\text{tr} \Sigma_g(\tau)}.$$

$$\mathbf{x}^* = \frac{\mathbf{x} - \mu_g}{s_g(\tau)}.$$

Each region and lag has its own factor. Shape and angle are preserved; absolute displacement amplitude is removed.

Thesis Fig. 3.21, all three rows. Right: eigenvalue ratio; labels give axis angles.

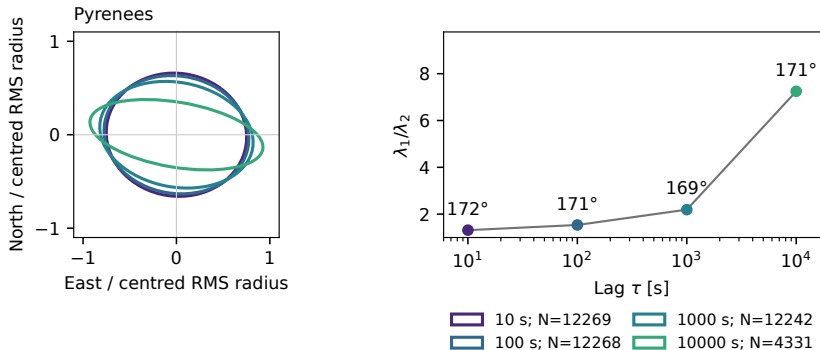
# Alps: the principal axes change with lag



At 1000 s:  $\lambda_1/\lambda_2 = 2.13$ , axis  $65.7^\circ$ ; 90,737 flights.

Thesis Fig. 3.21, original Alps row. Each ellipse uses its own centred RMS radius.

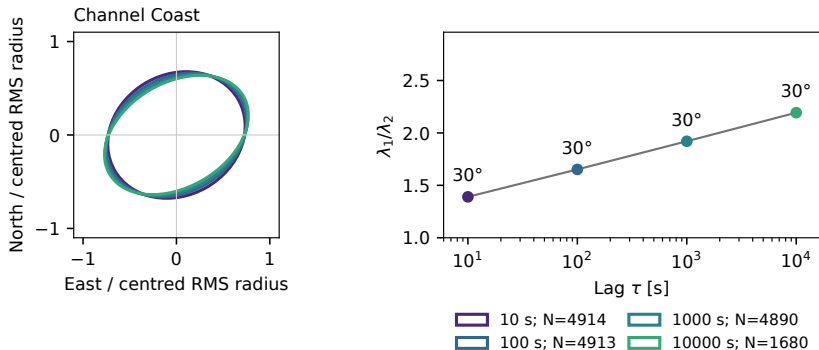
## Pyrenees: the principal axes change with lag



The axis approaches east–west and the eigenvalue ratio rises at long lag, while the contributing population falls.

Thesis Fig. 3.21, original Pyrenees row. Each ellipse uses its own centred RMS radius.

# Channel Coast: the principal axes change with lag



At 1000 s:  $\lambda_1/\lambda_2 = 1.92$ , axis  $29.7^\circ$ ; 4,890 flights.

Thesis Fig. 3.21, original Channel Coast row. Each ellipse uses its own centred RMS radius.

A rotation into the principal frame makes the components uncorrelated at the fitted lag and preserves radial lengths:

$$|U_g^\top \mathbf{X}| = |\mathbf{X}|.$$

Whitening with a reference covariance changes lengths:

$$\mathbf{Z}(\tau) = \Sigma_g(\tau_0)^{-1/2}[\mathbf{X}(\tau) - \boldsymbol{\mu}_g(\tau)].$$

Its covariance is the identity at  $\tau_0$ . At other lags, that remains a property to test. Whitening each lag separately would enforce equal principal variances in every ellipse.

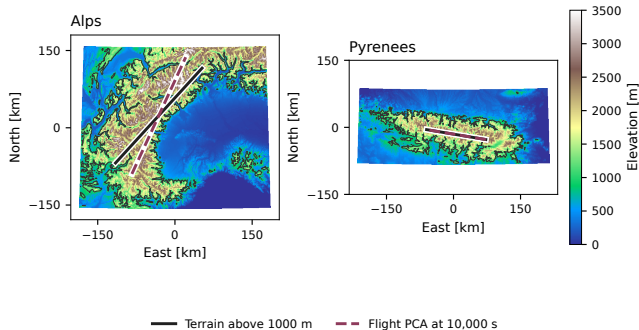
Thesis Section 3.5.2. Neither uncorrelated components nor an isotropic covariance establish a rotationally invariant law.

## The terrain axis comes from an independent elevation map

1. Crop NOAA ETOPO 2022 to the Alpine and Pyrenean take-off boxes.
2. Keep every fourth grid cell in both directions: one-arc-minute spacing.
3. Select cells at or above 1000 m; give higher cells no extra weight.
4. Project their centres to local WGS84 east/north coordinates at the box centre.
5. Weight each cell by  $\cos(\text{latitude})$  to approximate its area.
6. Compute the weighted centre and spatial covariance.
7. Take its leading eigenvector as the mean highland axis.

Thesis Section 3.5.2, Independent terrain orientation. Zero projection height; angles counterclockwise from east, modulo  $180^\circ$ .

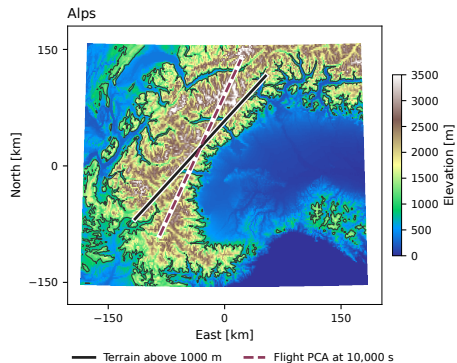
# Terrain and flight axes align more closely in the Pyrenees



Pyrenees: separation below  $1^\circ$ . Alps:  $17.6^\circ$ , with the flight axis more northward.

Thesis Fig. 3.22, both maps. Solid dark: terrain axis; dashed colour: flight PCA at 10,000 s. Line lengths carry no physical information.

# Alps: shared sector, with an approximately 18-degree offset



Terrain:  $47.6^\circ$ .

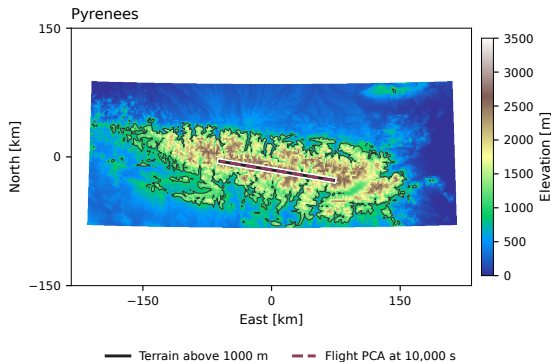
Flight PCA:  $65.2^\circ$ .

Separation:  $17.6^\circ$ .

46,336 flights support 10,000 s.

Black: highland-footprint axis. Dashed colour: current flight PCA, shifted to the same centre. Line lengths have no physical meaning.

# Pyrenees: the long-lag spread follows the highland axis



Terrain:  $170.5^\circ$ .

Flight PCA:  $170.5^\circ$ .

Separation: less than  $1^\circ$ .

4,331 flights support 10,000 s.

ETOPO footprint above 1000 m; threshold sensitivity:  $169.7\text{--}175.4^\circ$ . Decimal agreement does not imply this physical precision.

## The terrain comparison depends on the footprint definition

| Region   | Terrain axis | Flight axis | Separation |
|----------|--------------|-------------|------------|
| Alps     | 47.6°        | 65.2°       | 17.6°      |
| Pyrenees | 170.5°       | 170.5°      | <1°        |

Thresholds of 500, 1000 and 1500 m give:

Alps: 42.1°–47.9°,      Pyrenees: 169.7°–175.4°.

These spans measure sensitivity to the selected highland area. They are not confidence intervals. Decimal agreement does not imply that the physical mountain direction is known to that precision.

Thesis Section 3.5.2, commentary following Fig. 3.22.

## Same population as the 10,000-s PCA

- 1,680 paraglider flights.
- 2,013 nonoverlapping windows, wholly inside retained segments.
- Mean cleaned GNSS altitude along each window; equal window weights.

**Mean:** 1,006 m

Equal flight weights: 980 m.

Middle half of window means:  
854 m–1,160 m.

Approximate altitude above sea level;  
recorder datums are not harmonised.

Trapezoidal integration on the same 10-s segment grids. The target describes the supported long-lag windows; it is not take-off altitude or height above the terrain.

## ERA5 wind is interpolated to the measured flight altitude

**2016–2025, hourly; nine cells in the coastal box; 1000/925/850/700 hPa.**

$$H = \Phi / g_0, \quad h = \frac{R_{\oplus} H}{R_{\oplus} - H}, \quad \theta_w = \text{atan2}(\langle v(h) \rangle, \langle u(h) \rangle).$$

Convert geopotential to each hour's level heights. Interpolate **E/N wind components** to 1,006 m, between above-ground bounds.

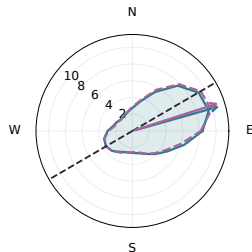
**Main comparison: all year, all hours vs. all year, 09–17 UTC.**

April–September is a separate seasonal check. Winds at 10/100 m AGL provide lower-level controls.

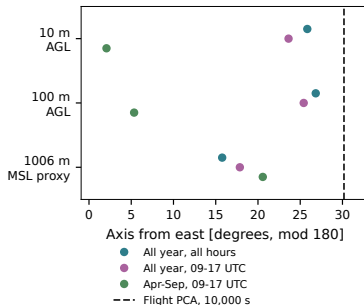
[ECMWF/Copernicus ERA5 pressure levels](#), accessed through a pinned snapshot of [Earthmover's public cloud-optimised ERA5 copy on AWS](#). Earthmover distributes this lossless copy; it is not a separate reanalysis. Complete hourly extract: 32.22 MB compressed, preserved in the repository and on SSD.

# At flight height, both annual references remain offset from PCA

At mean flight altitude: 1006 m  
Towards-sector frequency [%]



Height and time selection



Annual gaps at flight height:  $14.4^\circ$  for all hours,  $12.3^\circ$  for 09–17 UTC.

Thesis Fig. 3.23, complete figure. Angles counterclockwise from east. The rose shows where air moves towards at 1,006 m; the black line is the  $30.2^\circ$  centred PCA axis.

# Height and time checks preserve a broad orientation resemblance

| At 1,006 m          | Wind axis | PCA gap | $\rho_w$ |
|---------------------|-----------|---------|----------|
| All year, all hours | 15.8°     | 14.4°   | 0.43     |
| All year, 09–17 UTC | 17.9°     | 12.3°   | 0.44     |
| Apr–Sep, all hours  | 16.0°     | 14.2°   | 0.35     |
| Apr–Sep, 09–17 UTC  | 20.6°     | 9.6°    | 0.37     |

At the quartile heights, annual axes span 14.3–17.2° (all hours) and 16.4–19.3° (daytime), on shared support.

$\rho_w = \|\langle \mathbf{w} \rangle\| / \langle \|\mathbf{w}\| \rangle$  measures cancellation. Height ranges are sensitivities, not confidence intervals. The main estimates retain all selected hours.

## What the environmental comparisons establish

| Region        | Conclusion supported by this comparison                                                                                                    |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Pyrenees      | Close terrain–flight alignment; a geographical association worth testing along tracks.                                                     |
| Alps          | Same broad sector, with an approximately 18-degree offset from the mean highland axis.                                                     |
| Channel Coast | Wind at mean flight altitude lies about 12–14 degrees from PCA in both annual time selections; seasonal checks retain a broad resemblance. |

A common constant wind shifts the mean displacement by  $\tau\mathbf{w}$ ; centring removes it. Variable winds and route responses can still shape covariance.

Independent regional references constrain an interpretation. They do not separate wind, terrain, route choice and long-flight selection.

**MSL proxy:** the recorded GNSS altitude is treated approximately as altitude above mean sea level; receiver datums have not been harmonised.

**AGL:** 10 and 100 m above the ERA5 model surface. These are lower-level wind controls, with a different vertical reference.

$$\rho_w = \frac{\|\langle \mathbf{w} \rangle\|}{\langle \|\mathbf{w}\| \rangle}.$$

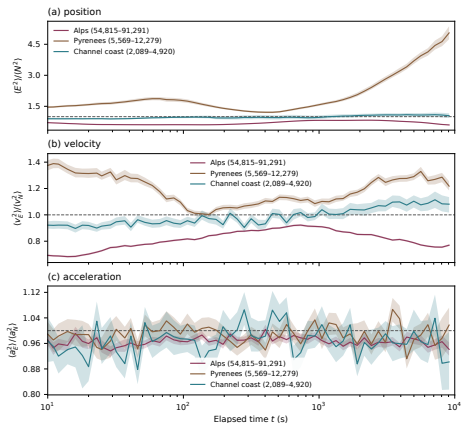
$\rho_w = 1$  means no directional cancellation; opposite winds can give a small resultant even when their individual speeds are large. Annual values of 0.43 and 0.44 indicate substantial cancellation.

Thesis Section 3.5.2, coastal wind reference.  $\rho_w$  is not an angular confidence level.

Thesis §3.5.3

**Terrain, equipment and wind**

# Position, velocity and acceleration have different imbalances



All three panels use raw east/north second moments at time since launch.

Median position / velocity ratios:

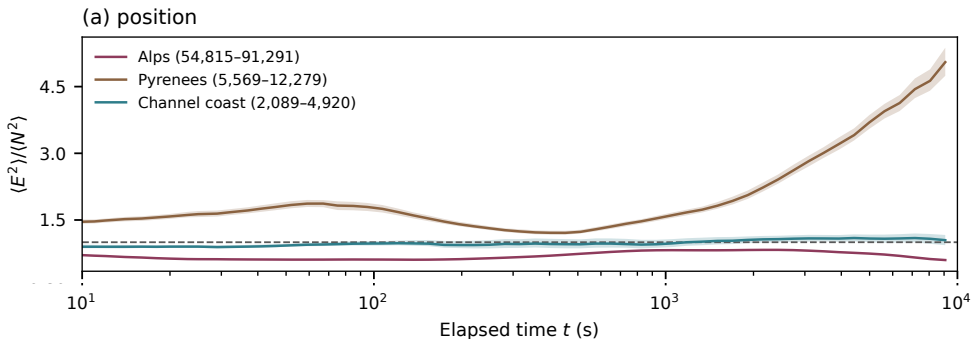
Alps: 0.67 / 0.81.

Pyrenees: 1.67 / 1.19.

The Pyrenean east–west preference is consistent with terrain influencing headings; wind and route selection remain possible contributors.

Thesis Fig. 3.24, all three panels, including acceleration. Shading: pointwise 10–90% site-day bootstrap ranges.

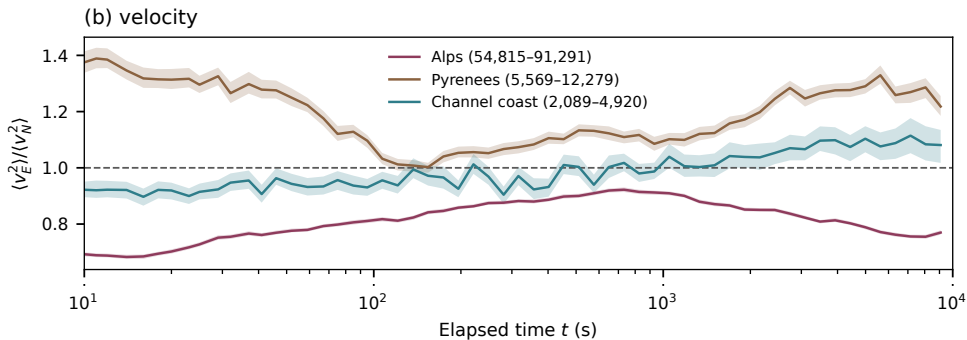
## Regional position moments differ along east and north



The Pyrenean position ratio rises above five at late times. The Alpine curve stays below one; the coast lies closer to one.

Thesis Fig. 3.24(a), enlarged original panel.

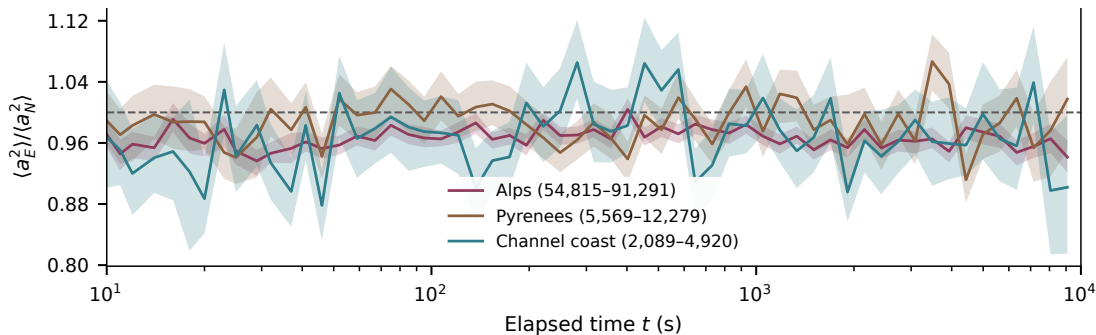
## Regional velocity moments show a related directional contrast



East–west preference is also visible in Pyrenean velocity moments. The relative spread varies less than the position ratio.

Thesis Fig. 3.24(b), enlarged original panel.

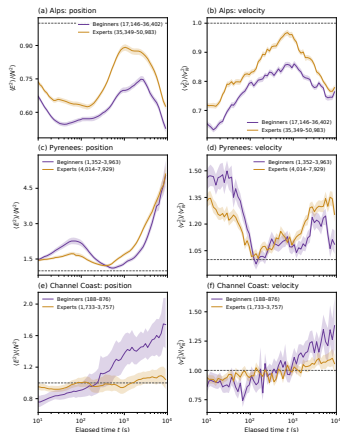
## Regional acceleration ratios are closer to unity



Acceleration has more balanced coordinate moments than the large position contrasts. Spatially varying wind can still contribute.

Thesis Fig. 3.24(c), original panel. This is a ratio of raw second moments.

# Equipment contrasts differ between the three regions



Thesis Fig. 3.25, all six panels. The next slides enlarge each original row.

Violet: EN A/B.

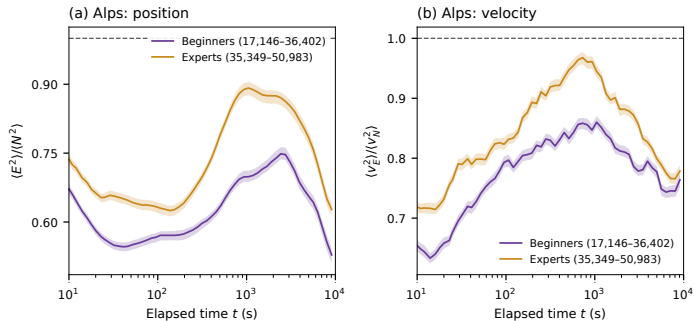
Ochre: EN C/D/CCC.

The legend uses these equipment groups as beginners/experts proxies.

Actual pilot skill is unmeasured.

Tandem, non-certified and unclassified records are excluded.

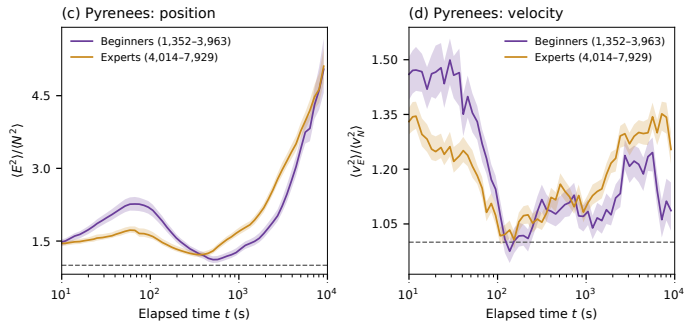
# Alps: C/D/CCC has smaller coordinate-moment imbalance



C/D/CCC stays closer to unity over the supported times. Median position ratio: 0.70, versus 0.61 for A/B.

Thesis Fig. 3.25, Alpine row.

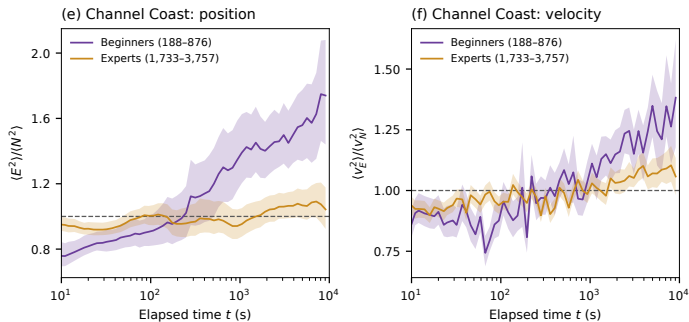
# Pyrenees: the ordering of the two groups reverses



Neither group remains uniformly closer to one. Near 9086 s, position ratios are about 5.05 for A/B and 5.11 for C/D/CCC.

Thesis Fig. 3.25, Pyrenean row.

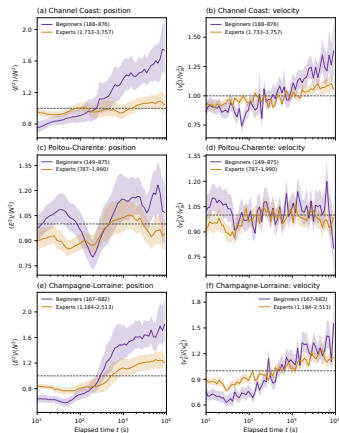
# Channel Coast: the late position contrast is pronounced



Near 9086 s: 1.74 for A/B and 1.04 for C/D/CCC, with 188 and 1733 contributing flights.

Thesis Fig. 3.25, coastal row.

## Two additional regions extend the coastal comparison

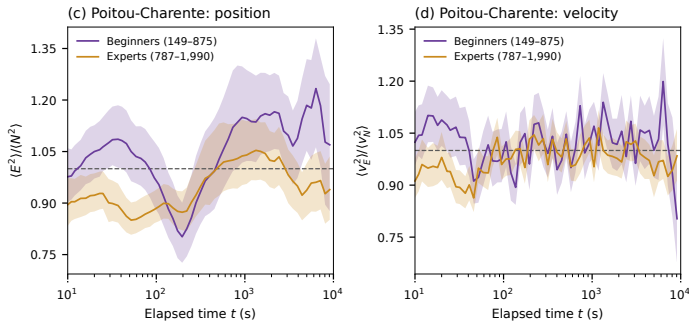


Thesis Fig. 3.26, all six panels.

The coastal row is repeated to compare it directly with Poitou-Charente and Champagne-Lorraine.

These boxes classify take-offs. They do not measure the terrain crossed by each flight or ensure that the equipment groups experienced comparable weather.

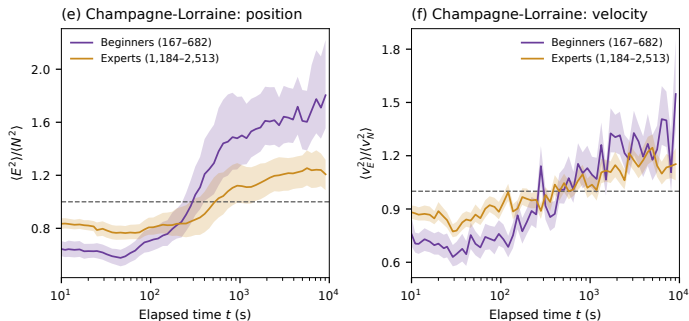
# Poitou-Charente: the group curves cross



The curves cross and their bands overlap broadly. A consistent equipment ordering is not supported.

Thesis Fig. 3.26, middle row.

# Champagne-Lorraine: a late contrast does not keep C/D/CCC at unity



A late position contrast appears, but the C/D/CCC ratio also rises above one.

Thesis Fig. 3.26, bottom row.

## **Hypothesis to discuss**

Equipment and pilot choices may moderate directional constraints associated with wind and terrain. Similar mountain profiles may reflect shared route constraints.

## **Evidence limiting that interpretation**

The Pyrenean ordering reverses; class is an experience proxy; coastal terrain is not absent; late support differs; and ensemble balance does not establish individual isotropy or performance.

## Equal east and north moments do not establish isotropy

$$\rho_{EN} = 1, \quad \Sigma = \begin{pmatrix} 1 & 0.9 \\ 0.9 & 1 \end{pmatrix}, \quad \lambda_1/\lambda_2 = 19.$$

This centred Gaussian has balanced coordinate moments but strongly unequal principal variances.

A pooled mixture of east-directed and north-directed routes can also balance the ensemble while each flight remains directional.

## Coastal flights still encounter terrain and selection effects

The Channel Coast includes cliffs and valleys, including the Pays de Caux. Wind and terrain may jointly orient lift and routes.

Equipment groups can differ in sites, weather, task and duration. Their bootstrap bands reflect one grouping structure; the groups have not been meteorologically matched.

Poitou-Charente and Champagne-Lorraine do not reproduce a uniform C/D/CCC advantage. The coastal observation alone does not establish a general flatland mechanism.

Thesis Section 3.5.3, Equipment, terrain and selection.

## Ground velocity combines air-relative motion and wind

$$\mathbf{v}_{\text{ground}} = \mathbf{v}_{\text{air}} + \mathbf{w}_h.$$

For a wind field  $\mathbf{w}_h(\mathbf{r}, z, t)$ ,

$$\mathbf{a}_{\text{ground}} = \frac{d\mathbf{v}_{\text{air}}}{dt} + \partial_t \mathbf{w}_h + (\mathbf{v}_{\text{ground}} \cdot \nabla_h) \mathbf{w}_h + \dot{z} \partial_z \mathbf{w}_h.$$

A constant additive wind disappears under differentiation. A steady wind field can still affect acceleration when the glider crosses horizontal or vertical gradients.

Thesis Section 3.5.3, Wind and ground velocity.

## Compare equipment groups under similar flight conditions

1. Restrict comparisons to shared take-off sites and days, with comparable flight duration and task.
2. Separate mean ground drift, centred spread and individual route directions.
3. Match wind to flight times and heights, and terrain to the actual tracks.
4. Where possible, compare the same pilot using different wings.

Upwind progress, air-relative performance and choice of route are different outcomes. The equipment interpretation depends on which one is being tested.

Thesis Section 3.5.3, Discriminating comparisons.